

Describe alg for Elliptic curve

$$P, Q \in E(\mathbb{F}_q)$$

Find d s.t. $Q = dP$

$\log_P Q = d$ if d is the least positive integer with $Q = dP$

Ex: $E = \{(x, y) \in \mathbb{F}_{27}^2 \mid y^2 = x^3 + 14x + 19\} \cup \{\emptyset\}$

$P = (57, 58) \ \& \ Q = (56, 48)$ Find $\log_P Q$

$P, Q \in E?$

We can use Shanks Baby step Giant-step algorithm.

Suppose $Q = kP$ for $k \geq 0$ [have to find the least k if exists]

Find the least integer ^{m} greater than $\sqrt{n}$ ($n = \#(E)$)

(?) $k = mg + r$ for $0 \leq r < m$

$B = \{(Q - rP, r) \mid 0 \leq r < m\}$ (hg^{-r}, r)

check if we get

$Q - jP = \infty$

stop if you get $0 \leq j \leq m$

$$\Rightarrow \frac{Q - jP = \infty}{\text{so that } \log_p Q = j}$$

If this step fails then go to the
final step.

$$mP, 2mP, 3mP, \dots, jmP, \dots$$

find j s.t. $\underline{j m P = Q - r P}$ (compare with members of B)

$$\text{Then } \underline{Q = (r + jm)P}$$

Ex. $E = \{(x, y) \in \mathbb{F}_{11}^2 \mid y^2 = x^3 + 3x + 1\} \cup \{0\}$

$P = (1, 4)$ & $Q = (3, 2)$. Find $\log_P Q$

In this case $E = C_{18}$. (i.e. cyclic gp of order 18)

In a change table $m = 9$ (instead of $\sqrt{18} = 5 \geq \sqrt{18}$) least.

$$\begin{aligned} B &= \{(Q, 0), (Q - P, 1), (Q - 2P, 2), \dots, (Q - 8P, 8)\} \\ &= \{(3, 2), 0), (5, 3), 1), (6, 9), 2), (9, 3), 3), \underline{(4, 0), 4),} \\ &\quad (9, 8), 5), (16, 4), 6), (15, 8), 7), (13, 9), 8)\} \end{aligned}$$

observe $(\infty, j) \notin B$. So go to the final step

Compute $mP, 2mP, \dots$

Hence $\boxed{P = (1, 4), m = 9}$, $9P = (4, 0)$.

~~order $(4, 0) = 2$~~

$$q=1, r=4, m=9, \quad k = qm + r = 1 \times 9 + 4 = 13$$

so that $\log_p Q = \underline{13}$

Ex: Find $\log_p Q$ taking $m=5$

Pollard's ρ algorithm.

To find $\log_p Q$ for $P, Q \in G$

$$G = G_1 \cup G_2 \cup G_3$$

$$E = G_1 \cup G_2 \cup G_3$$

$$G_i \cap G_j = \emptyset \text{ for } i \neq j.$$

$$f: E \rightarrow E$$

$$f(R) = \begin{cases} P + R & \text{if } R \in G_1 \\ 2R & \text{if } R \in G_2 \\ Q + R & \text{if } R \in G_3 \end{cases}$$

Choose $m_0 \in \{1, 2, \dots, n\}$. Define $R_0 \equiv m_0 P$
 $R_1 = f(R_0)$ (i.e. $R_{i+1} = f(R_i)$)

$$= \begin{cases} P + m_0 P = \underline{(1+m_0)P} & \text{if } R_0 \in G_1 \\ \underline{2m_0 P} & \text{if } R_0 \in G_2 \\ \underline{m_0 P + Q} & \text{if } R_0 \in G_3 \end{cases}$$

Set $R_i = m_i P + n_i Q$.

$$R_{i+1} = m_{i+1} P + n_{i+1} Q$$

$$\text{where } m_{i+1} = \begin{cases} m_i + 1 & \text{if } R_i \in G_1 \\ 2m_i & \text{if } R_i \in G_2 \\ m_i & \text{if } R_i \in G_3 \end{cases}$$

$$n_{i+1} = \begin{cases} n_i & \text{if } R_i \in G_1 \\ 2n_i & \text{if } R_i \in G_2 \\ n_i + 1 & \text{if } R_i \in G_3 \end{cases}$$

If we get $m_i P + n_i Q = m_{i+j} P + n_{i+j} Q$

If $Q = kP$, then

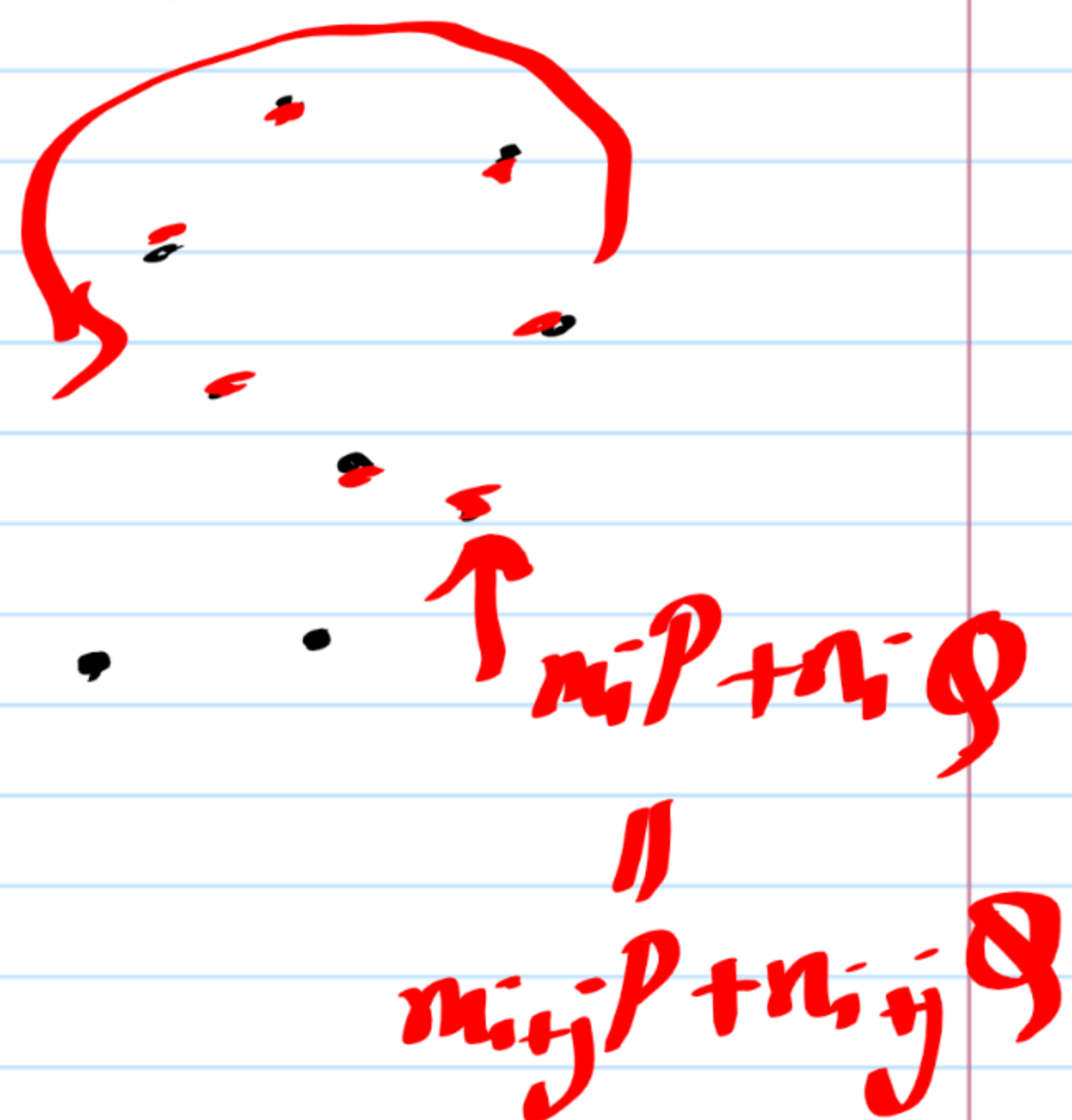
$$m_i P + n_i k P = m_{i+j} P + n_{i+j} k P$$

Find k

$$k = \frac{m_i - m_{i+j}}{(n_{i+j} - n_i)} \pmod{n} \quad \text{if } (n_{i+j} - n_i)^{-1} \text{ exists in } \mathbb{Z}_n.$$

otherwise find $\gcd(n_{i+j} - n_i, n) = d$

Do modulo $\frac{n}{d}$?



~~$$k = n_{i+j} - n_i$$~~

$$k = (n_{i+j} - n_i)^{-1} (n_i - n_{i+j}) \pmod{\frac{n}{d}}$$

~~E~~
Ex: $E = \{(x, y) \in \mathbb{F}_{11}^2 \mid y^2 \equiv x^3 + 3x + 1\} \cup \{\infty\}$

$$P = (1, 4), Q = (3, 2)$$

$$E = \{(0, 1), (0, 10), (1, 4), (1, 7), (2, 2), (2, 9)\} \cup$$

$\parallel$
 G_1

$$\cup \{(3, 2), (3, 9), (4, 10), (5, 3), (5, 8), (6, 2)\} \cup$$

$\parallel$
 G_2

$$\{(6, 9), (8, 3), (8, 8), (9, 3), (9, 8), \infty\}$$

$\parallel$
 G_3

$$E = G_1 \cup G_2 \cup G_3$$

Pick, $n_0 = 2, n_0 = 0, R_0 = 2P = 2(1, 4) = (2, 9)$

$$f: E \rightarrow E \quad \nearrow \quad R_0 = (2, 9) \in G_1$$

$$R_1 = \underbrace{(1, 4) + (2, 9)} = \underbrace{(0, 1)} \in G_1$$

$$R_2 = \underbrace{(1, 4) + (0, 1)} = \underbrace{(8, 8)} \in G_3$$

$$R_3 = Q + R_2 = (3, 2) + (8, 8) = (1, 7) \in G_1$$

$$R_4 = P + R_3 = (1, 4) + (1, 7) = \infty \in G_3$$

$$R_5 = Q + R_4 = (3, 2) + \infty = (3, 2) \in G_2$$

$$R_6 = 2R_5 = 2(3, 2) = (9, 8) \in G_3$$

$$R_7 = Q + R_6 = (3, 2) + (9, 8) = (0, 1) = R_1$$

$$\Rightarrow m_1 P + n_1 Q = m_2 P + n_2 Q$$

$$\Rightarrow h = \frac{m_1 - m_2}{(n_2 - n_1)} \pmod{18}$$

Find m_i, n_i

$$m_1 = m_0 + 1 = 3, \quad n_1 = n_0 = 0 \quad \text{as } R_0 \in G_1$$

$$m_2 = m_1 + 1 = 3 + 1 = 4$$

since $R_1 \in G_1$

$$n_2 = n_1 = 0$$

$$\cancel{m_2 = m_1 + 1 = 3 + 1 = 4} \quad m_3 = m_2 = 4$$

since $R_2 \in G_3$

$$\cancel{n_2 = n_1 = 0}$$

$$n_3 = n_2 + 1 = 0 + 1$$

$$m_7 = m_6 = 10$$

since $R_6 \in G_3$

$$n_7 = n_6 + 1 = 4 + 1 = 5$$

$$(3 - 10)P = (5 - 0)Q$$

$$\Rightarrow Q = \frac{-5^{-1} \times 7}{\pmod{18}}$$

$$= -10 \times 7$$

$$= \underline{13}$$

Gur3 12th Nov